

KELT-23Ab

R-2795

Benchmark run · workflow W8-benchmark-deep · record deep-bench_KELT-23A_260731 · created 2026-09-04T23:03:24+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461252.6853 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1649 +/- 0.0098
Transit depth (Rp/Rs)^2	2.72 +/- 0.32 [%]
Orbital Inclination (inc)	89.91 +/- 0.86
Airmass coefficient 1 (a1)	33.53 +/- 9.53
Airmass coefficient 2 (a2)	-2.34 +/- 0.16
Scatter in the residuals of the lightcurve fit is	23.03 %
Best Comparison Star	#3 - [510, 337]
Optimal Aperture	6.19
Optimal Annulus	22.6
Transit Duration (day)	0.0838 +/- 0.0012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1649 $\pm$ 0.0098 vs 0.132 (deviation 0.29 σ)
Duration fit vs published	0.0838 d vs 0.0954 d (deviation 0.84 σ)
Mid-time O-C	2.0 min
Depth SNR	1.5
Residual scatter	23.03 %

Light curve

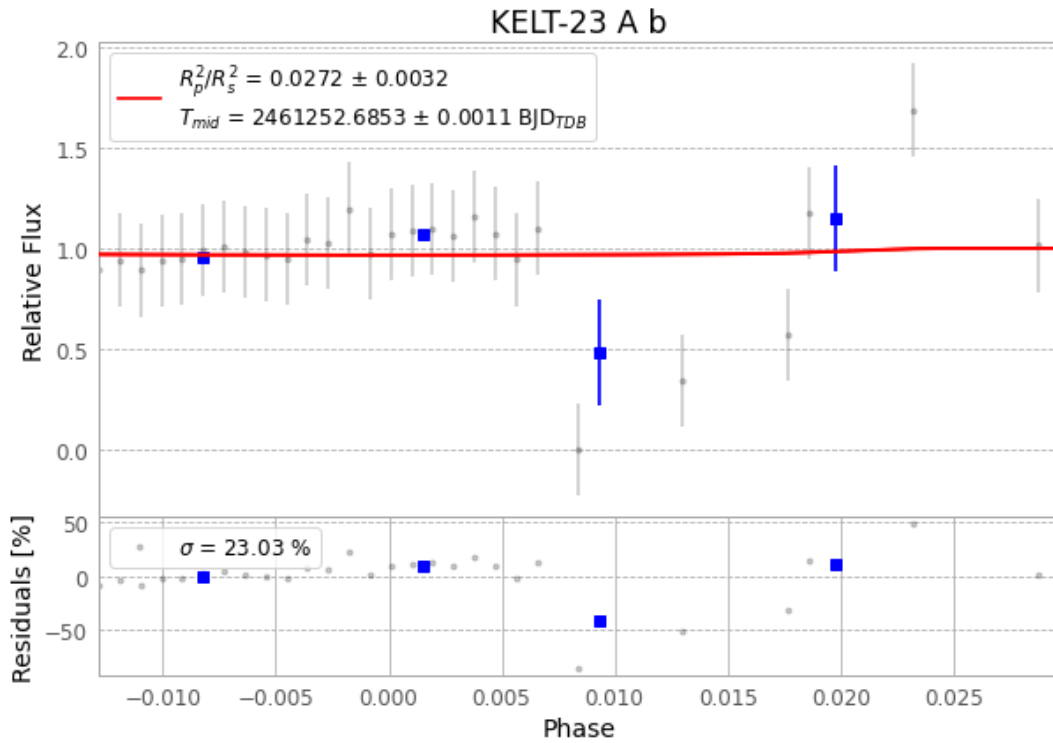


Figure 1 — FinalLightCurve_KELT-23 A b_2026-07-30.png (SHA-256 45de500c8c451013...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [302, 167], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 df170ebc6ba0babc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

47 FITS frames (31 MB), KELT-23A260731034517.FITS → KELT-23A260731060315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KELT-23A-260731.log (fc7a8f934e8e...), FinalLightCurve_KELT-23 A b_2026-07-30.png (45de500c8c45...), FinalLightCurve_KELT-23 A b_2026-07-30.csv (a782f7b6673f...)

Compute resources

Wall time 491.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 23:03Z.